7.4 Trapezoids and Kites

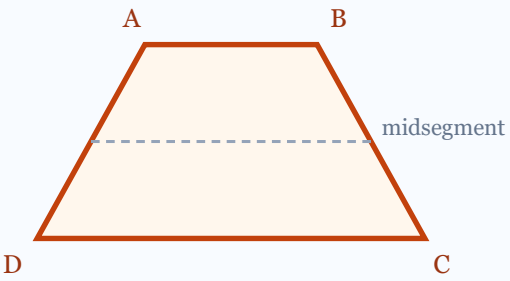
A **trapezoid** has exactly one pair of parallel sides, called the bases. Its **midsegment** (median) connects the midpoints of the two legs, runs parallel to both bases, and has length equal to the average of the two base lengths: $m = \dfrac{b_1 + b_2}{2}$. An **isosceles trapezoid** (congruent legs) has an extra pair of properties: both base angles at each base are congruent to each other, and the diagonals are congruent. A **kite** has two distinct pairs of consecutive congruent sides (not opposite, like a parallelogram). Its diagonals are always perpendicular; the diagonal connecting the vertices between the congruent sides is the axis of symmetry — it bisects the other diagonal *and* bisects the two vertex angles it passes through, while the two angles between the unequal sides are congruent to each other. This worksheet uses all of these relationships together.

Examples

Example 1: Midsegment from two bases

A trapezoid has bases 10 and 16. Find the length of the midsegment.

Solution



The midsegment joins the midpoints of the legs and is parallel to both bases.

Identify the two bases: $b_1 = 10$, $b_2 = 16$.

The midsegment is the average of the two bases:

$$m = \frac{b_1 + b_2}{2} = \frac{10 + 16}{2} = \frac{26}{2} = 13.$$

Answer: 13.

Example 2: Find a missing base

A trapezoid has midsegment 14 and one base 9. Find the other base.

Solution

Identify the known midsegment and base: $m = 14$, $b_1 = 9$, and set up the midsegment equation with $b_2 = b$ unknown.

Solve for the missing base:

$$\frac{9 + b}{2} = 14$$

$$9 + b = 28$$

$$b = 19.$$

Answer: 19.

Example 3: Isosceles trapezoid base angles

An isosceles trapezoid has base angles of 70° at its longer base. Find the two angles at the shorter base.

Solution

In an isosceles trapezoid, both base angles at the same base are congruent, and each pair of angles sharing a leg is supplementary (co-interior angles between the parallel bases).

Angle at the shorter base $= 180^\circ - 70^\circ = 110^\circ$.

By symmetry, both angles at the shorter base equal 110°.

Answer: 110° and 110°.

Example 4: Isosceles trapezoid diagonals are congruent

In isosceles trapezoid $ABCD$, diagonal $AC = 5x - 2$ and diagonal $BD = 3x + 8$. Find x and the length of each diagonal.

Solution

Diagonals of an isosceles trapezoid are congruent: $AC = BD$.

$$5x - 2 = 3x + 8$$

$$2x = 10 \Rightarrow x = 5.$$

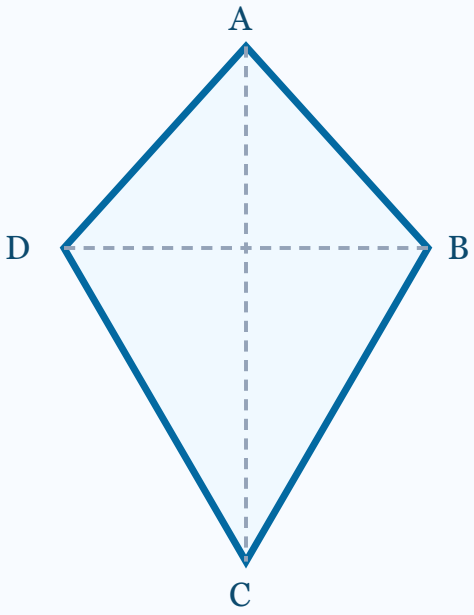
$$AC = 5(5) - 2 = 23.$$

Answer: $x = 5$; each diagonal $= 23$.

Example 5: Kite angle sum

In kite $ABCD$ (with $AB = AD$ and $CB = CD$), the two congruent non-vertex angles are $m\angle B = m\angle D = 100°$, and $m\angle A = 40°$. Find $m\angle C$.

Solution



$AB = AD$ and $CB = CD$; the axis $\overline{AC}$ splits the kite into two congruent triangles.

All four angles of any quadrilateral sum to $360°$:

$40 + 100 + 100 + m\angle C = 360$

$240 + m\angle C = 360$

$m\angle C = 120°$.

Answer: $120°$.

Example 6: Kite area from diagonals

A kite has diagonals 12 and 16. Find its area.

Solution

A kite's diagonals are perpendicular, so its area is half the product of the diagonals — just like a rhombus:

$A = \frac{1}{2}(12)(16) = \frac{1}{2}(192) = 96.$

Answer: 96.

Example 7: Isosceles trapezoid legs

An isosceles trapezoid has legs $3x + 2$ and $x + 10$. Find x and the length of each leg.

Solution

The legs of an isosceles trapezoid are congruent:

$3x + 2 = x + 10$

$2x = 8 \Rightarrow x = 4.$

Each leg $= 3(4) + 2 = 14.$

Answer: $x = 4$; each leg $= 14$.

Example 8: Kite perimeter

Kite $ABCD$ has $AB = AD = 8$ and $CB = CD = 5$. Find the perimeter.

Solution

Identify the two pairs of consecutive congruent sides: $AB = AD = 8$ and $CB = CD = 5$.

Add all four sides:

Perimeter $= AB + BC + CD + DA = 8 + 5 + 5 + 8 = 26.$

Answer: 26.

Example 9: Trapezoid consecutive angles along a leg

In trapezoid $ABCD$, $\overline{AB} \parallel \overline{DC}$, and $m\angle A = 95°$. Find $m\angle D$ (the angle sharing leg $\overline{AD}$ with $\angle A$).

Solution

$\angle A$ and $\angle D$ are co-interior (same-side interior) angles formed by transversal $\overline{AD}$ across the parallel bases, so they are supplementary:

$m\angle D = 180° - 95° = 85°$.

Answer: $85°$.

Example 10: Application — window brace

A trapezoidal window frame has a horizontal brace connecting the midpoints of its two slanted sides (legs). The bottom base is 34 in, and the brace measures 24 in. Find the length of the top base.

Solution

The brace is the midsegment of the trapezoid, so it equals the average of the two bases:

$$\frac{34 + t}{2} = 24$$

$$34 + t = 48$$

$$t = 14.$$

Answer: the top base is 14 in.

Example 11: Kite axis bisects the vertex angle

In kite $ABCD$ with $AB = AD$ and $CB = CD$, diagonal $\overline{AC}$ is the axis of symmetry. If $m\angle A = 68°$, find $m\angle BAC$.

Solution

The axis of symmetry $\overline{AC}$ bisects the vertex angles it passes through (here, $\angle A$ and $\angle C$), splitting each into two congruent parts:

$$m\angle BAC = \frac{68°}{2} = 34°.$$

Answer: $34°$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Midsegment

Bases 12 and 20. Midsegment?

Answer: **Verification:**

Problem 2: Base

Midsegment 11, base 8. Other base?

Answer: **Verification:**

Problem 3: Isosceles

Base angles 65°, 65°. Find the other two base angles.

Answer: **Verification:**

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Problem 4: Kite area

Diagonals 10 and 14. Area?

Answer: **Verification:**

Problem 5: Midsegment

Bases 5 and 17. Midsegment?

Answer: **Verification:**

Problem 6: Consecutive

$\overline{AB} \parallel \overline{DC}$; $m\angle B = 100^\circ$. Find $m\angle C$ (shares leg $\overline{BC}$).

Answer: **Verification:**

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Problem 7: Kite perimeter

Sides 9, 9, 6, 6. Perimeter?

Answer: **Verification:**

Problem 8: Legs

Isosceles trapezoid; legs $4x - 3$ and $2x + 7$. Find x and each leg.

Answer: **Verification:**

Problem 9: Base

Midsegment 18, base 22. Other base?

Answer: **Verification:**

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Problem 10: Kite property

Diagonals of a kite are always...

Answer: **Verification:**

Problem 11: Isosceles angles

One base angle is 78°. Find the other angle at the *same* base.

Answer: **Verification:**

Problem 12: Kite area

Diagonals 9 and 20. Area?

Answer: **Verification:**

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Problem 13: Midsegment formula

Bases a and b. Write the midsegment formula.

Answer: **Verification:**

Problem 14: Challenge

Isosceles trapezoid: diagonal $AC = 4x - 6$ and diagonal $BD = 2x + 10$. Find x and the length of each diagonal.

Answer: **Verification:**

Problem 15: Challenge

Kite $ABCD$ has $m\angle B = m\angle D = 110°$ (the non-vertex angles). If $m\angle A = 3x$ and $m\angle C = x$, find x and both vertex angles.

Answer: **Verification:**

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✓ Answer Key

1. 16
2. 14
3. 115°, 115°
4. 70
5. 11
6. 80°
7. 30
8. $x = 5$; legs = 17
9. 14
10. Perpendicular
11. 78° (base angles at the same base are congruent)
12. 90
13. $m = \dfrac{a+b}{2}$
14. $x = 8$; each diagonal = 26
15. $x = 35°$; $m\angle A = 105°$, $m\angle C = 35°$

Great work — keep practising!